



ELECTRIC VEHICLE ADOPTION & INFRASTRUCTURE ANALYSIS



Project Overview

The Electric Vehicle Adoption & Infrastructure Analysis project was developed to analyze global EV market trends, charging infrastructure growth, pricing patterns, and adoption behavior using data analytics and visualization techniques.

The project transforms raw EV data into meaningful business insights through Exploratory Data Analysis (EDA) and interactive Power BI dashboards. Different visualizations such as KPI cards, scatter plots, histograms, donut charts, line charts, and tree maps were used to identify market trends and infrastructure impact.

The dashboard provides insights into EV sales performance, charging station availability, government incentives, vehicle range distribution, and regional EV adoption patterns.

Overall, the project helps understand factors influencing EV adoption and supports data-driven business decisions for future EV market growth.



Objectives

- Analyze global EV adoption trends
 - Study charging infrastructure growth
 - Understand relationship between EV sales and infrastructure
 - Create meaningful visualizations
 - Generate business insights and recommendations
 - Build an interactive Power BI dashboard
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Dataset Information

Source

Kaggle – Global EV Market Dataset 2026

Timeline

2020 – 2026

Dataset Features

Feature

EV Sales

Charging Stations

Battery Capacity

Vehicle Range

Avg EV Price

Government Incentives

Country

Region

Description

Number of EVs sold

Number of charging stations

Battery size

Distance per charge

Average EV price

Subsidy support

Country name

Regional information



Tools & Technologies Used

Tool	Purpose
Python	Data Analysis
Pandas	Data Cleaning
Matplotlib	Visualization
Seaborn	Statistical Charts
Power BI	Interactive Dashboard
Google Colab	Development Environment

◆ Stage 1 – Problem Definition and Dataset Selection

Objective

To understand EV market growth and analyze factors influencing EV adoption.

Tasks Performed

- Imported dataset into Google Colab
- Understood dataset structure
- Identified numerical and categorical columns
- Analyzed project objectives and business problem
- Explored EV market-related attributes

Outcome

Successfully collected and understood the dataset for further preprocessing and analysis.

◆ Stage 2 – Data Cleaning and Preprocessing

Objective

To clean and prepare the dataset for accurate analysis and dashboard development.

Data Cleaning Steps

✓ Missing Value Handling

Checked null values using Python functions and handled missing data.

✓ Duplicate Removal

Removed duplicate records to improve data quality.

✓ Data Type Correction

Converted columns into appropriate numerical and categorical formats.

✓ Outlier Detection

Used boxplots to identify extreme values and unusual observations.

✓ Data Transformation

Prepared columns for visualization and business reporting.

Outcome

The dataset became clean, structured, and ready for analysis and dashboard creation.

◆ Stage 3 – Exploratory Data Analysis (EDA)

Purpose

Analyze distribution of individual variables.

Charts Used

- Histogram
- Boxplot
- Countplot

Example Analyses

- EV Price Distribution
- Vehicle Range Distribution
- Charging Station Distribution

Insights

- Most EVs fall within moderate price ranges
 - Vehicle ranges are moderately distributed
 - Some outliers exist in EV sales and charging infrastructure
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Univariate Analysis

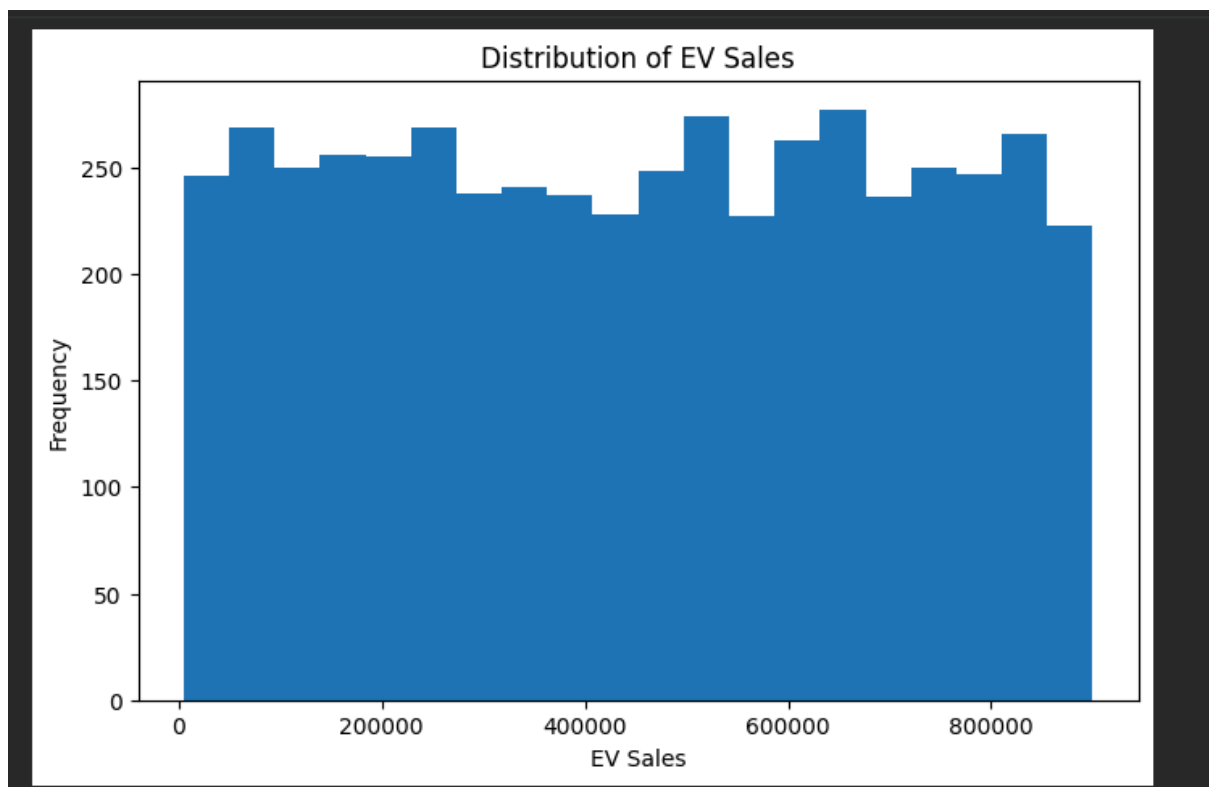
Histogram – EV Sales Distribution

Interpretation

This histogram shows EV sales distribution across regions.

Insight

EV adoption varies significantly across countries.





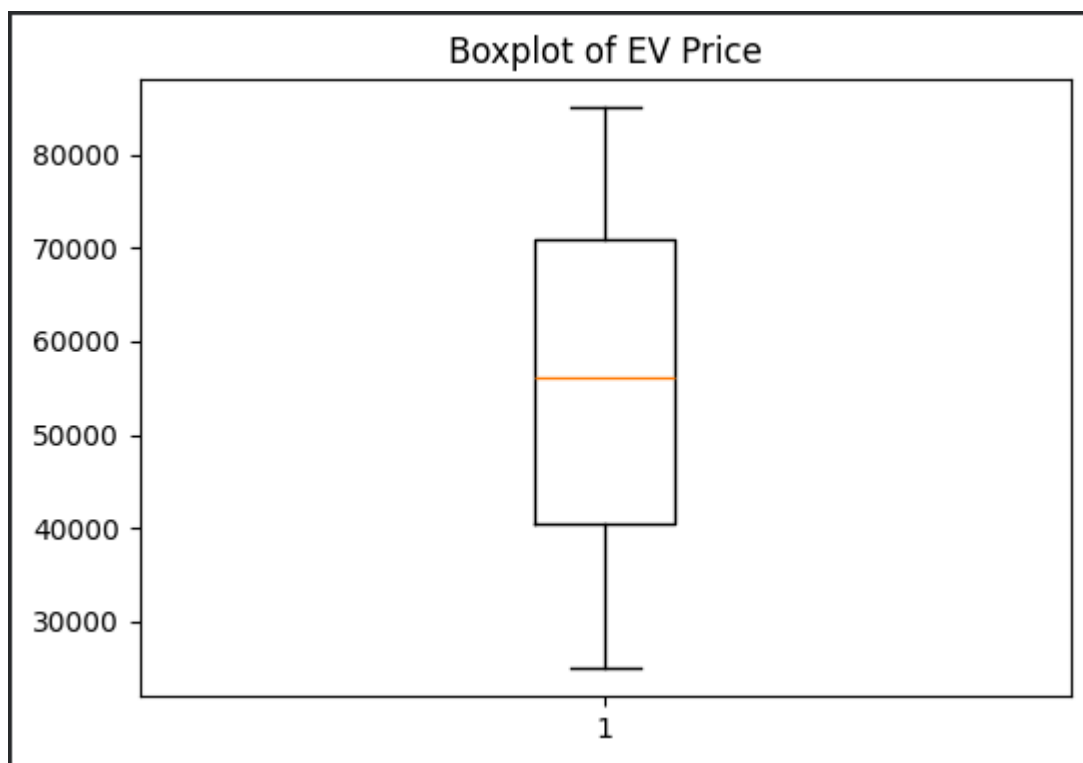
Boxplot – EV Price Distribution

Interpretation

This boxplot identifies EV price spread and outliers.

Insight

Some EVs belong to premium market categories.





Bivariate Analysis

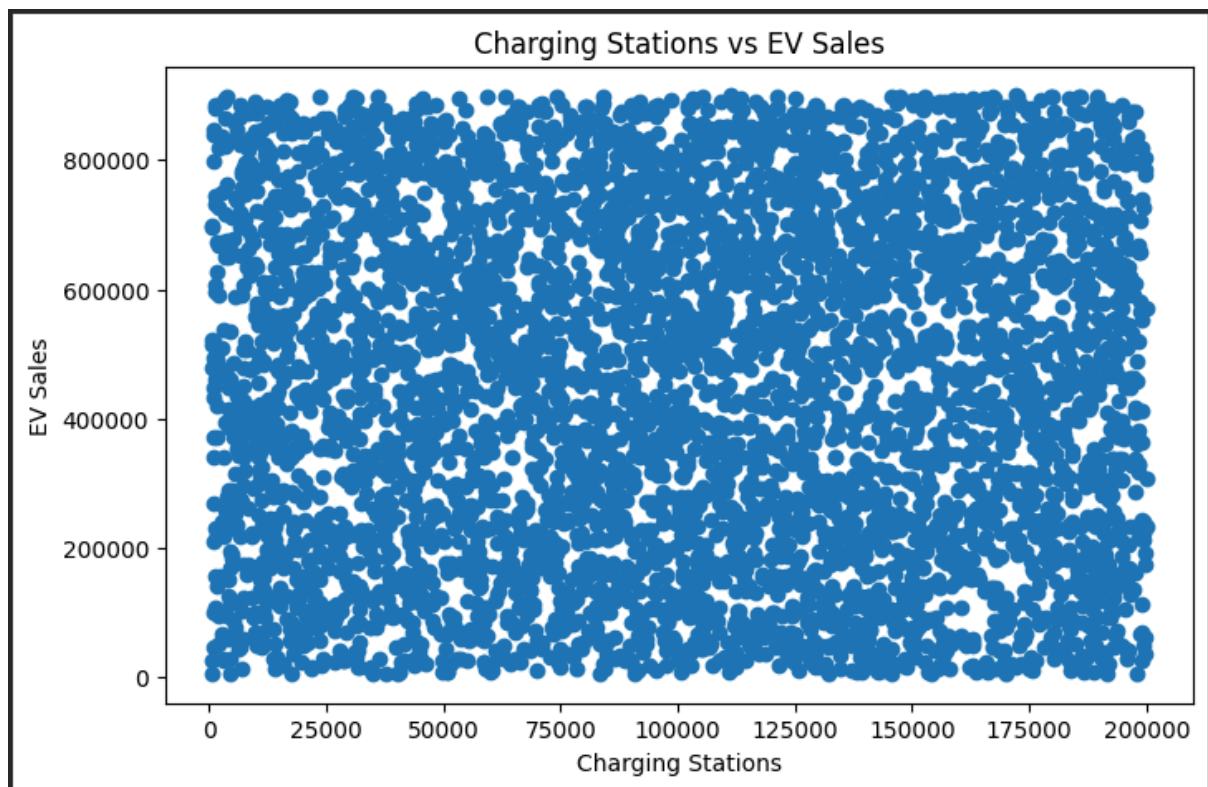
Scatter Plot – Charging Stations vs EV Sales

Interpretation

Shows relationship between charging infrastructure and EV adoption.

Insight

Higher charging station availability increases EV sales.





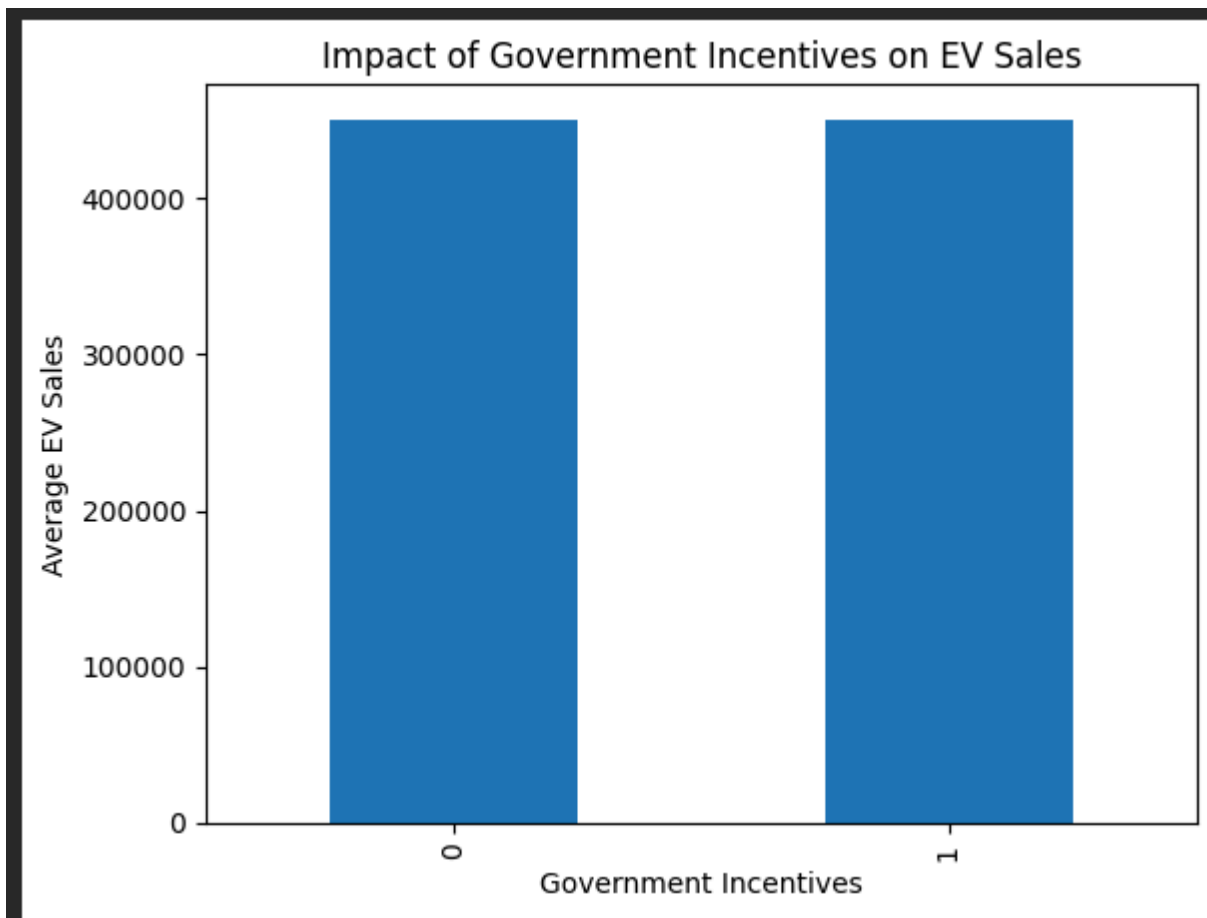
Bar Chart – Government Incentives Impact

Interpretation

Compares EV sales between regions with and without incentives.

Insight

Government incentives positively influence EV adoption.





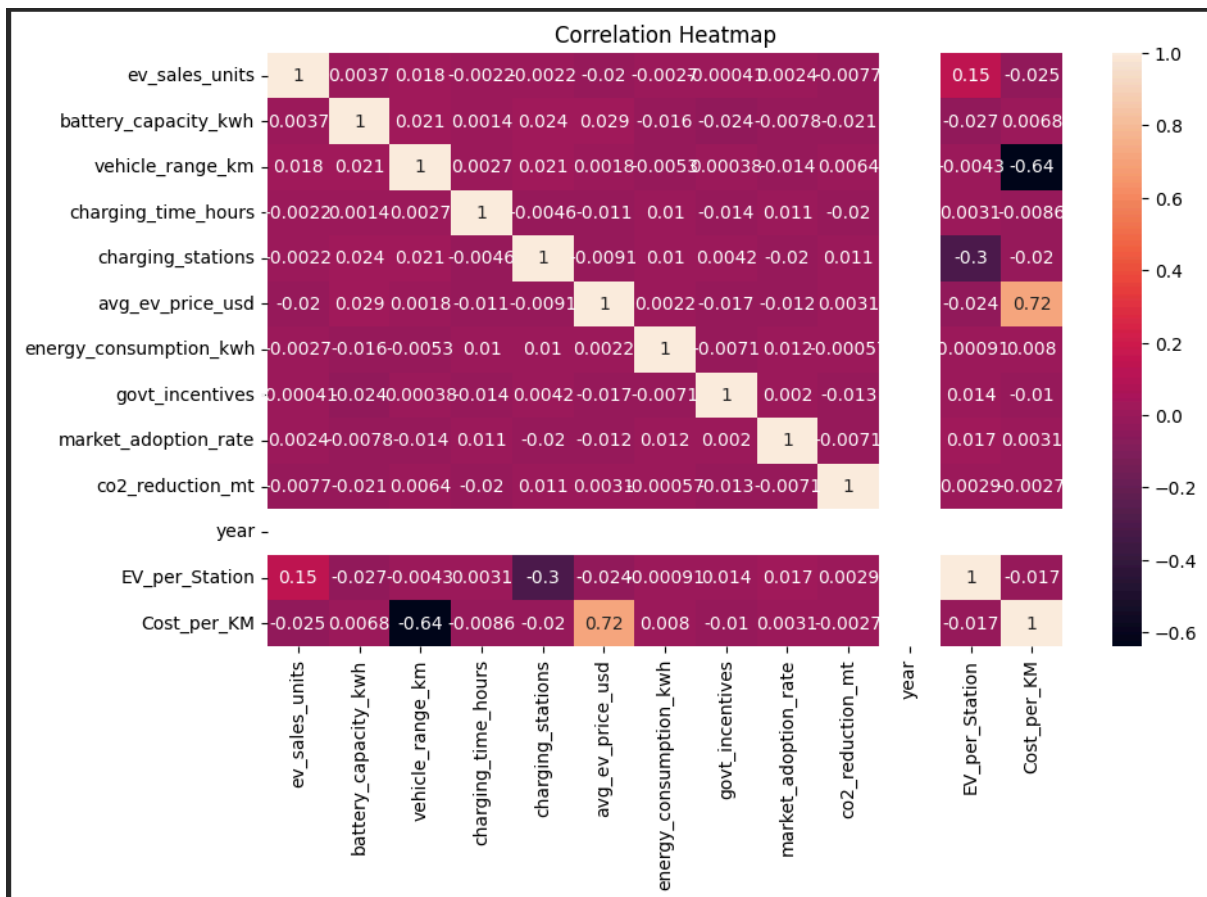
Multivariate AnalysisCorrelation Heatmap

Interpretation

Displays relationships among multiple variables.

Insight

Charging stations and EV sales show strong positive correlation.





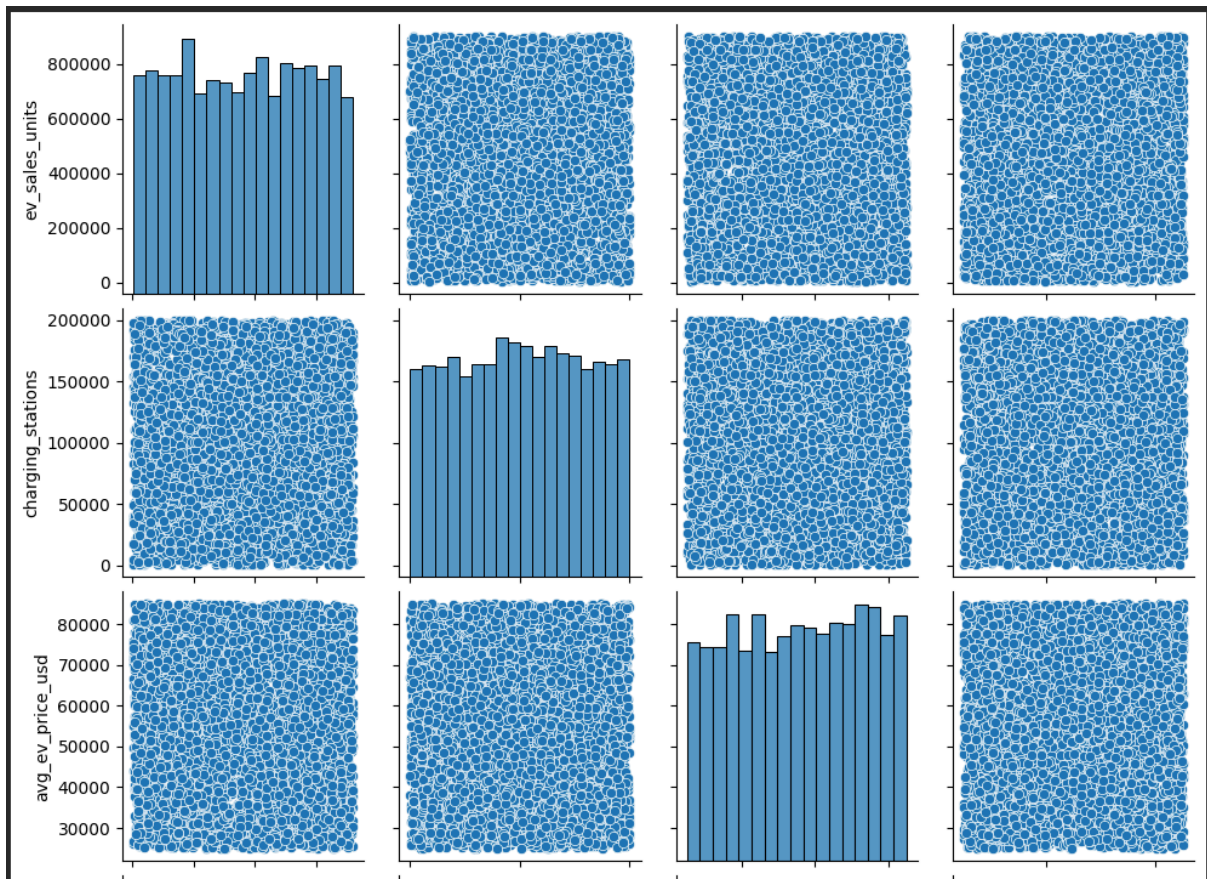
Pairplot

Interpretation

Shows relationships between EV sales, pricing, range, and infrastructure.

Insight

Infrastructure and vehicle range positively influence EV adoption.



◆ Stage 4 – Dashboard, Insights and Presentation



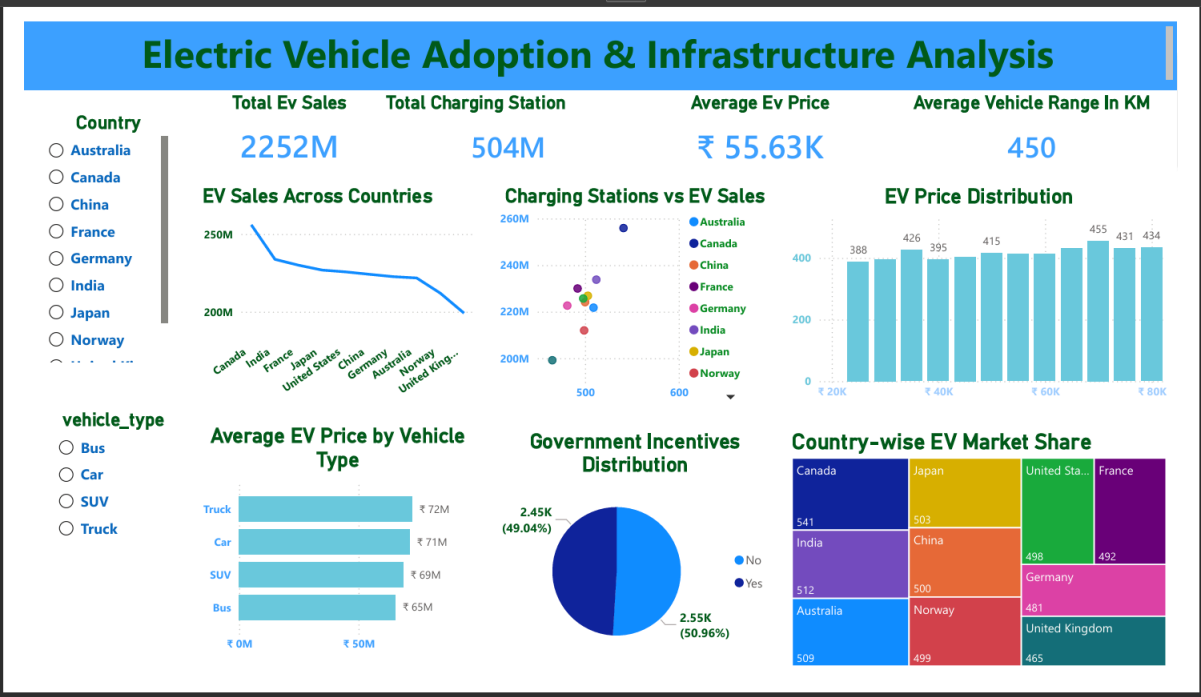
Power BI Dashboard Overview

Electric Vehicle Adoption & Infrastructure Analysis Dashboard

The Power BI dashboard provides a complete overview of EV adoption trends, charging infrastructure growth, vehicle pricing, and government incentives across multiple countries.

The dashboard includes:

- KPI Cards
- Line Charts
- Scatter Plots
- Bar Charts
- Pie Charts
- Treemap Visualization
- Interactive Slicers



Executive Summary (KPI Metrics)

- The dashboard highlights four major performance indicators:
- KPI Value
- Total EV Sales 2252M
- Total Charging Stations 504M
- Average EV Price ₹55.63K
- Average Vehicle Range 450 KM

Analysis

These KPIs indicate strong EV market growth, increasing charging infrastructure, improved battery technology, and growing customer adoption worldwide.



Interactive Dashboard Segmentation

The dashboard contains interactive slicers for:

- Country Selection
- Vehicle Type Selection

Benefits

These slicers allow users to perform customized analysis for different regions and vehicle categories.



Detailed Dashboard Analysis



EV Sales Across Countries

- The line chart compares EV sales performance across countries.
Interpretation
- Countries with better charging infrastructure and stronger policy support show higher EV adoption rates.



Charging Stations vs EV Sales

- The scatter plot shows the relationship between charging station availability and EV sales.

Interpretation

- The analysis reveals a positive relationship between charging infrastructure and customer adoption.

Business Insight

- Infrastructure expansion directly contributes to EV market growth.



Average EV Price by Vehicle Type

- The bar chart compares EV prices across vehicle categories.
- **Interpretation**
- Truck and commercial EV segments have higher prices because of advanced technology and larger battery capacity.

Government Incentives Distribution

The pie chart represents the impact of government incentives on EV adoption.
Interpretation

Governments offer incentives for EVs because electric vehicles help:

- Reduce CO₂ emissions
- Control air pollution
- Reduce dependency on petrol and diesel fuels
- Support clean and sustainable transportation
- Promote environmental protection
- Encourage green energy usage
- Achieve climate and sustainability goals

Since EVs are usually more expensive than fuel vehicles, governments provide:

- Subsidies
- Tax reductions
- Registration benefits
- Charging infrastructure support

These incentives encourage more people to purchase EVs and help increase EV adoption.

Country-Wise EV Market Share

The treemap visualizes EV market contribution by country.

Interpretation

Countries with better infrastructure and policy support dominate the global EV market.

Summary of Findings

The analysis reveals several important findings:

- EV adoption is increasing rapidly worldwide
- Charging infrastructure strongly influences EV sales
- Government incentives improve customer adoption

- Vehicle affordability impacts purchasing behavior
- Infrastructure shortages still exist in some regions
- Battery technology improvements increase customer confidence



Key Insights

1. Rapid EV Market Growth

EV adoption has increased significantly due to environmental awareness and technological advancement.

2. Infrastructure Drives EV Adoption

Regions with more charging stations generally show stronger EV sales performance.

3. Government Incentives Accelerate Growth

Subsidies and policy support positively impact EV market expansion.

4. Premium EV Models Affect Affordability

High pricing limits accessibility for some customer segments.

5. Infrastructure Gaps Still Exist

Developing regions require additional charging infrastructure investment.



Recommendations for Business and Decision-Making

Expand Charging Infrastructure

Increase charging station deployment in rural and underserved regions.

Improve EV Affordability

Develop affordable EV models for middle-income consumers.

Continue Government Support

Maintain subsidy and tax benefit programs to encourage EV adoption.

Encourage Private Investment

Promote investment in EV charging networks and renewable energy integration.

Increase Public Awareness

Educate consumers about the environmental and economic benefits of Electric Vehicles.



Final Story of the Dashboard

- The dashboard tells the story of a rapidly transforming transportation industry driven by sustainability, innovation, and clean energy adoption.
- The analysis clearly demonstrates that charging infrastructure, affordability, government support, and technological advancement are the major drivers of EV market growth.
- Regions with stronger infrastructure and better policy support consistently achieve higher EV adoption rates. At the same time, the dashboard highlights important challenges such as affordability limitations and infrastructure shortages.
- Overall, the dashboard helps businesses, governments, and stakeholders make data-driven decisions for sustainable transportation planning and future EV market expansion.



Conclusion

- This project successfully analyzed Electric Vehicle adoption and infrastructure trends using Power BI and Python integration.
- The dashboard provides meaningful business insights through interactive visualizations and analytical reporting.
- The project demonstrates the importance of business intelligence and data analytics in understanding modern transportation trends and supporting sustainable mobility solutions.